

R1

<u>sid</u>	<u>bid</u>	<u>day</u>
22	101	10/10/96
58	103	11/12/96

S1

<u>sid</u>	sname	rating	age
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

B1

<u>bid</u>	<u>bname</u>	<u>color</u>
101	tiger	red
103	lion	green
105	hero	blue

S2

<u>sid</u>	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

Q1: $\pi_{sid,sname,age}(\sigma_{rating < 10}(S_2))$

Q2: $\pi_{sname, rating}(\sigma_{S_1.sid > R_1.sid}(S_1 \times R_1))$

Q3: $\pi_{sid,bname}(\sigma_{bid}(R_1 \times B_1))$

Q4: $S_2 / (\pi_{sid,age}(S_2) / \pi_{sid}(\sigma_{bid=103}(R_1)))$

Q5: $(S_2 \bowtie_* R_1) \cup (B_1 \bowtie_* R_1)$

Q6: $\pi_{sname, rating, age, bname, color}(((S_1 \cup S_2) \bowtie_{sid} R_1) \bowtie_{bid} \sigma_{color=red}(B_1))$

Q7: $\pi_{sid}(S_2) - \pi_{sid}(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating > 7}(S_1)) - \pi_{sid,sname}(S_2))$

Q8: Conver the query of Q1 to DRC formula and TRC formula.

Q9: Conver the query of Q2 to DRC formula and TRC formula.

Q10: Conver the query of Q3 to DRC formula and TRC formula.

Q11: What's data model? What's data schema?

Q12: Students(sid: string, name: string, ID number: string, cid: string)

Courses(cid: string, cname:string)

List all super keys, candidate keys, primary keys, foreign keys of table Students.

Q13: What are the relational operations on the relational data model? What are their differences and similarities?